

F-65 Megawing - SensorAndTrackEngine Engineering Model

Phase-3 Version 1.0

Status: Human-reviewed engineering baseline / approved rewrite input

24 August 2026

This version is the human-reviewed SensorAndTrackEngine baseline for the next coordinated rewrites of Architecture Invariants, Gameplay & Simulation Requirements, Engine Runtime, Read-First alignment, and the generated Phase-3 schemas. It does not silently supersede an already approved parent document. Where v1.0 intentionally changes a current parent invariant or player-visible contract, the required propagation is stated explicitly and must be carried into the applicable rewrite / configuration-control record.

Human review is complete for the engineering model. No unresolved architecture-interpretation question remains inside this document. Numeric tables, schedules, thresholds, exact binary layouts, measured byte totals, and tuning values remain intentionally Phase-3 gated.

Primary parent set for rewrite alignment: Architecture Invariants 1.5.1; Engine Runtime 0.2; Gameplay & Simulation Requirements; Physics v3.3; Graphics v2.1; Technical Alignment / Read-First.

1. Sensor / Weapon Ownership Boundary

SensorAndTrackEngine owns observation and assessment. WeaponAndDamageEngine owns authoritative missile state and physical weapon evolution.

SensorAndTrackEngine:

- evaluates radar and IR observables
- evaluates chaff / flare competition against sensor and seeker candidates
- generates SeekerAssessment records
- generates radar-emission observations that may be received by RWR
- owns semantic tactical-track processing and tactical-link fusion within the applicable knowledge domain

WeaponAndDamageEngine:

- owns WeaponGuidanceState
- owns missile seeker-lock / target / countermeasure state
- owns support / autonomy transitions
- applies guidance commands at stage 12
- owns missile 3DOF motion, propulsion, physical guidance integration, fuze resolution, detonation, damage, and physical lifecycle

SensorAndTrackEngine decides what a seeker can observe. It does not directly mutate the missile's authoritative WeaponGuidanceState. A SeekerAssessment produced at stage 15 becomes consumable by WeaponAndDamageEngine no earlier than stage 12 of the following simulation tick. No same-tick seeker-to-guidance feedback is permitted.

Countermeasure entities remain the existing chaff / flare entities on the frozen weapon/countermeasure physical path.

2. Cross-Module Read-Only Views

SensorAndTrack never reaches into WeaponAndDamage private arrays. It receives bounded read-only projections after weapon/countermeasure integration.

2.1 SeekerInputView (logical)

- missile EntityHandle
- seeker profile ID
- seeker field of view / pointing state
- last legitimate supported target estimate
- current candidate-lock state required for retention evaluation
- seeker operating / capability flags

2.2 CountermeasureView (logical)

- decoy EntityHandle and owner EntityHandle
- decoy class: CHAFF or FLARE
- position / velocity
- release tick / age
- maneuver-qualified state
- decoy profile ID

When an active missile exposes an EmitterView, that view is also a projection from Weapon-owned missile / seeker state. SensorAndTrack does not acquire ownership of the missile merely because it evaluates the seeker radar.

3. Cross-Tick Handoff Storage and Byte Ownership

RadarObservation may be pure stage-15 scratch. Interfaces that cross a tick boundary use fixed persistent handoff storage.

Interface	Direction	Required storage
LinkTrackReport	Tick N stage 15 -> later stage 15	Bounded scheduled link-report queue
SeekerAssessment	Tick N stage 15 -> Tick N+1 stage 12	One pending slot per guided-missile slot
WeaponSupportUpdate	Tick N stage 15 -> Tick N+1 stage 12	One pending slot per guided-missile slot

Human-reviewed ownership decision: these cross-tick records belong to a fixed SensorHandoffState owned and byte-accounted by SensorAndTrackEngine. SensorHandoffState contains the 32 SeekerAssessment mailboxes, 32 WeaponSupportUpdate mailboxes, and the bounded tactical-link pending queue. WeaponAndDamage consumes the missile mailboxes read-only at stage 12; SensorAndTrack may overwrite them only according to the generated source-tick / supersession rule after the legal consumer point.

The mailboxes and link queue are not free memory. Their exact sizes, alignment, padding, and total owner cost close in the generated schema / measured-limits rewrite. No dynamic allocation is permitted.

4. Tick Placement and Causality Contract

SensorAndTrackEngine executes exclusively at stage 15.

```
Stage 11-14 Weapons acceptance, integration, damage resolution
Stage 15     SensorAndTrackEngine
Stage 16     AI / RIO consume completed stage-15 state
...
Stage 19     PresentationExtractor -> immutable PresentationSnapshot
```

One-tick causality (normative):

- Weapon requests accepted at stage 11 use sensor / track state committed on the previous tick.
- Missile support evaluated at stage 12 cannot depend on an observation, link report, WeaponSupportUpdate, or SeekerAssessment generated at stage 15 of the same tick.
- AI / RIO may issue commands no earlier than the tick following the sensor state they consume.
- A MEGA LINK / RED LINK report generated at stage 15 becomes eligible for recipient ingestion no earlier than the following simulation tick.
- A WeaponSupportUpdate produced from stage-15 fire-control state on tick N is available to missile guidance no earlier than stage 12 of tick N+1.
- A SeekerAssessment produced at stage 15 on tick N is consumable by WeaponAndDamageEngine no earlier than stage 12 of tick N+1.

5. Capacity and Knowledge-Domain Contract

Pool / state	Capacity	Budget / note
Radar truth contacts	32	32 x 48 B; shared physical truth-contact candidate set
Semantic radar tracks	24 per tactical-link knowledge domain	80 B max each; MEGA and RED domains are capacity-isolated
Current active track domains	2	MEGA / BLUE and RED; 48 physical track records max
RIO priority tracks	4	Subset of MEGA-domain tracks only
Chaff / flare entities	64 shared	Existing pool; six live decoys per aircraft
Guided missiles	32	Existing pool; 16 simultaneous stress case

5.1 TRACK-DOMAIN-001 - Human-Reviewed Resolution

Resolution: the 24-track limit applies per active tactical-network knowledge domain, not globally across opposing sides. The current product has exactly two tactical knowledge domains: MEGA / BLUE and RED. Neutral is an identification state, not a third tactical-network domain.

Required parent rewrite: Architecture Invariants shall replace the ambiguous single global radar-track interpretation with an equivalent of two capacity-isolated 24-entry semantic track stores (or another generated representation that proves the same semantics). At the current 80-byte ceiling this is a maximum of 48 physical records / 3,840 bytes, an increase of 1,920 bytes over the current single-24 allocation before alignment or ancillary state. That delta must be charged to the measured SensorTrack owner ledger.

- Association, fusion, identity, capacity pressure, eviction, and track aging are strictly intra-domain.
- A Blue/MEGA track can never be evicted, degraded, or denied allocation because of private RED-domain track load, and vice versa.
- The same physical truth contact may have one semantic representation in each opposing domain because those records represent different knowledge. Within a domain, correlated organic/offboard information produces one fused semantic track.
- RIO priority tracks apply only to the MEGA / player tactical picture. RED doctrine priority and link transmit priority are independent concepts.
- No third Neutral track store is created. Neutral / friendly / hostile remain IdentificationState values inside a domain.

This is an intentional architecture rewrite requirement accepted by human review. It is not a claim that Architecture Invariants 1.5.1 already states these semantics.

5.2 Missile and Decoy Capacity Isolation

Active-radar and passive-IR missile seekers maintain missile-local seeker state through the WeaponAndDamageEngine contract. They do not allocate RadarTrack, radar-truth, or RIO-priority-track records. Sixteen simultaneously active missile seekers therefore consume zero semantic RadarTrack and zero RIO-priority-track slots.

Chaff does not allocate persistent RadarTrack or radar-truth slots. It is evaluated as a bounded decoy competitor using the existing chaff/flare entity pool.

6. Canonical Types and Sensor Identity

```
#include "f65_entity.h"    /* EntityHandle = canonical 4-byte handle */
#include "f65_world.h"     /* WorldPosition */
#include "f65_time.h"      /* SimulationTick */

typedef struct {
    EntityHandle owner;
    u8 sensor_instance_id;
} SensorRef;
```

One physical entity may own multiple radar instances, for example ship search plus fire-control or SAM acquisition plus engagement. Maximum instances per entity are schema- and profile-bounded. No new physical entity pool is created. RadarSensorState is authoritative persistent state owned by the platform/system record. EmitterView is its stage-15 projection. The same persistent-versus-projection rule applies to link state and RWR receiver history.

7. Orthogonal Classification

Side, platform class, radar role, sensor profile, and tactical-link membership are independent properties. Combination types such as BLUE_FIGHTER_RADAR are forbidden.

Property	Initial values / meaning
Side / affiliation	BLUE RED NEUTRAL / NONE
Platform class	Airborne fighter; Airborne AIC; Fixed surface; Mobile surface; Ship; Missile seeker
Radar role	Wide-area search; Acquisition; Multirole / fire-control; Dedicated fire-control; Active seeker; authored extensions
sensor_profile_id	Authoritative technical-performance profile
network_id	LINK_NET_MEGA LINK_NET_RED LINK_NET_NONE

8. Core Logical Records

Exact layouts close through the generated registry. Required logical records are:

- EmitterView - stage-15 projection; uses SensorRef.
- LinkNodeView - stage-15 projection; independent TX/RX capability and state.
- RadarObservation - stage-15 scratch sensor-visible measurement.
- ObservationProvenance - diagnostic/oracle only; truth_source is forbidden from association, AI, RIO, fire control, fusion, capacity scoring, and gameplay.
- LinkTrackReport - deliberately limited tactical report; quantized position permitted; contact class, identification state/authority, source tick, and minimal correlation metadata.
- RadarTrack - <= 80 B; lifecycle, quality, source/fusion, domain/network ownership, position/velocity, identity/confidence, support, priority, age/hits, error surrogate, rule-profile ID, capacity score, stable tie-breaks, last-update tick.
- RWR Emission Observation - ephemeral stage-15 received-emission observation.
- RWRContactState - bounded persistent receiver memory owned by the protected platform/system; public RWRContact is a projection.

- MissileThreatState - launch indication, active seeker, indication lost, still-dangerous state, and confirmed-defeat confidence.
- SeekerInputView / CountermeasureView - read-only WeaponAndDamage projections.
- SeekerAssessment - SensorAndTrack -> WeaponAndDamage handoff.
- WeaponSupportUpdate - SensorAndTrack -> WeaponAndDamage support handoff.

The generated registry shall define the canonical mapping between internal TrackLifecycle + quality/support/priority fields and the parent Gameplay-facing TrackQuality logical contract.

9. MEGA LINK / RED LINK Contract and Fan-Out Bound

MEGA LINK is the Blue/player tactical track-sharing network. RED LINK is its adversary-side counterpart. Eligible fighter, AIC, fixed/mobile surface, SAM, and ship nodes may transmit and receive according to authored LinkNodeView capability. Tactical link is separate from radar emission and separate from missile Weapon Support Datalink.

LinkTrackReport is intentionally less capable than a local fire-control track. It may provide position (including altitude when available), contact class, identification / interrogation result and authority, source, source tick, and bounded correlation metadata. It does not carry remote weapons-quality authority, remote SUPPORT_VALID, a complete fire-control solution, raw truth handles, or hidden truth velocity.

An OFFBOARD-only report can cue acquisition and populate the tactical display before ownship radar detection, but it cannot establish ownship weapons quality or ownship missile support. Organic ownship correlation is required.

Link workload must be capacity-proven. The generated schema shall define LINK_REPORTS_PER_NODE_PER_UPDATE_MAX and LINK_PENDING_REPORTS_MAX. When more reports are eligible than can be sent, deterministic selection uses link_tx_priority / report_priority_class, then confidence, freshness, and stable track ID. Link transmit priority is independent from the four RIO priority tracks.

Minimum link causality is one simulation tick. Exact cadence, range/geometry, LOS requirement, quantization, and additional transport latency remain Phase-3 gated.

9.1 Ownship Defensive Jammer and MEGA LINK - Approved Gameplay Rewrite Requirement

For the initial production behavior, while the F-65 defensive jammer is active:

- organic ownship radar is blanked;
- support relying on that non-autonomous organic radar is hard-invalidated at the next legal support-consumption point;
- MEGA LINK transmit is denied;
- MEGA LINK receive is denied.

Link reports that arrive while RX is denied are not buffered for later replay; they are lost. Previously received offboard tracks remain and age/coast normally. Remote recipients likewise retain and coast their previously received F-65 reports. When the jammer is disabled, radar and link resume only through their deterministic scan/update schedules; stale tracks are not snapped to perfect current state and no denied-link backlog is replayed.

External hostile jamming may degrade or deny MEGA LINK / RED LINK according to Phase-3-gated tables.

10. Track Lifecycle, Quality, and Eviction

Lifecycle: PLOT -> TENTATIVE -> TRACK -> COAST -> DROPPED
Quality: LOW -> FIRM -> WEAPON

Lifecycle and quality are independent. WEAPON quality enables a valid fire-control solution, support path, high-confidence targeting data, and RIO recommendation. It is not a launch-permission bit; the player may still release poor, unsupported, unidentified, friendly, neutral, or unauthorized shots subject to the separate weapon/ROE rules.

Eviction is intra-domain only. If a domain track store is full, the only eligible victims are non-priority COAST tracks in that same domain. Select the lowest capacity_score, then the frozen stable tie-break sequence. Priority or active tracks are not silently displaced. If no eligible victim exists, the generated overflow/fault contract determines the rejected creation/update result.

11. Simplified IR Signature Model

IR signature depends only on aircraft/signature profile, aggregate actual engine-emission state, and target aspect relative to the seeker. This is an approved gameplay rewrite requirement; exact bins and values remain Phase-3 data.

Power bins: IR_POWER_LOW | IR_POWER_MIL | IR_POWER_MAX. They derive from actual engine operating state, not raw pilot throttle command.

For multi-engine aircraft, individual engine states are deterministically mapped through the aircraft IR profile to one aggregate power bin or signature index. Engine-out and asymmetric-power cases therefore change signature without introducing a thermal engine model.

Target aspect is measured from the target longitudinal axis to seeker line of sight: 0 degrees = nose, 180 degrees = tail. Initial classes are IR_ASPECT_NOSE_BLIND, IR_ASPECT_FORWARD_QUARTER, IR_ASPECT_BEAM_AFT, and IR_ASPECT_TAIL. Nose / near-nose is blind or extremely weak; forward quarter is low; beam/aft is intermediate; tail is highest. Exact angular boundaries remain Phase-3 gated.

```
IR_signature_class = IRSignatureTable[target_profile][power_bin][aspect_bin]
acquirable = range <= IRDetectionRangeTable[seeker_profile][IR_signature_class]
```

Seeker angle off boresight affects field-of-view eligibility only. It does not independently alter the target IR-signature class. Skin-temperature, exhaust-fluid, solar-heating, and other thermal-physics models are out of scope.

12. Passive IR Seeker

A heater is passive. It creates no radar emitter and no RWR active-seeker indication. The F-65 has no dedicated IR missile-approach-warning system.

FOV -> range -> target aspect -> IR signature -> flare competition -> retention logic -> SeekerAssessment

The seeker acquires or retains the strongest valid candidate according to deterministic scoring and retention hysteresis. No probability-only flare-success roll is permitted.

13. Countermeasure Maneuver Qualification and Behavior

Approved gameplay rewrite requirement: flare and chaff require at least +3.0 G of actual achieved positive normal load during the approved release window to become effective seeker-seduction candidates.

```
CM_MANEUVER_QUALIFIED =  
    achieved_normal_load >= +3.0 G  
    during approved deterministic release window
```

Commanded stick / G demand is not used. Negative 3 G does not qualify. Exact filtering/window duration remains Phase-3 gated. If qualification fails, inventory is still consumed when the release is otherwise legal, the decoy entity still exists and is presented normally, but its seeker-seduction contribution is ineffective. Pulling G does not modify the aircraft base radar or IR signature; it qualifies decoy separation/effectiveness.

13.1 Flare

A flare competes with the target aircraft inside an IR seeker valid acquisition region. Deterministic score depends on flare IR profile, FOV, geometry, age/decay, target IR signature, angular separation, and maneuver qualification. A flare does not become a semantic radar track or tactical-link report.

13.2 Chaff

Chaff competes with the aircraft return for airborne, surface, ship, SAM, and active-missile radar processing. Deterministic score depends on chaff radar profile, FOV/gate geometry, Doppler compatibility, age/decay, aircraft RCS/aspect, clutter/notch condition, and maneuver qualification.

Chaff does not reduce intrinsic aircraft RCS and does not allocate persistent RadarTrack or radar-truth slots. It may deterministically degrade/reject an observation, induce mis-correlation, or drive an existing track toward coast. Persistent false tactical tracks and cross-target chaff seduction remain outside current scope.

13.3 Seeker Retention Hysteresis

```
retain current seeker candidate  
unless competitor_score > current_score + switch_margin  
for required dwell
```

Exact switch margin and dwell are Phase-3 gated. The rule prevents 100 Hz oscillation between near-equal aircraft/decoy candidates.

14. Radar-Defense Mechanisms - Geometry / Physics Driven

Do not create named BEAM_BONUS, NOTCH_BONUS, WEAVE_BONUS, or equivalent magic modifiers.

- Beaming - geometry drives radial velocity toward the clutter-rejection region; it does not automatically break track.
- Notching - succeeds only when beam geometry, relevant land/sea clutter, Doppler rejection, and sufficient dwell coincide.
- Weaving - changes heading, radial velocity, line of sight, predicted state, filter residual, confidence, and notch dwell; there is no arbitrary weave deception bonus.
- Chaff - provides a competing radar return as defined in Section 13.
- Jamming - modifies detection / track quality through deterministic authored tables.
- Terrain masking - authoritative world-query LOS break; rendered LOD never controls radar LOS.

The same conceptual radar-defense evaluator applies to fighter, AIC, fixed/mobile surface, SAM, shipborne, and active missile-seeker radars. Differences belong in sensor_profile_id, PRF/clutter tables, FOV, scan/revisit schedules, range tables, coast/reacquisition rules, jammer susceptibility, and chaff susceptibility.

15. Weapon Support Datalink and Active Radar Seeker

MEGA LINK / RED LINK are platform-to-platform tactical track-sharing networks. Weapon Support Datalink is the shooter/fire-control-to-missile support path. They are distinct systems.

The current long- and medium-range F-65 radar missiles are active-radar missiles with supported midcourse guidance. The pre-active phase is not SEMI_ACTIVE.

```
RAIL -> SUPPORTED -> AUTONOMOUS / ACTIVE_SEARCH -> TRACKING -> COASTING_REACQUIRING -> DEFEATED /  
DETONATED / EXPIRED
```

15.1 WeaponSupportUpdate

WeaponSupportUpdate is produced only from a valid committed local fire-control / support track. It carries the supported missile handle, supporting SensorRef, legitimate target estimate (position / velocity available from the local track), source tick, support state, and any required activation cue. It never contains a raw truth target handle or information unavailable to the supporting fire-control track. Update cadence is deterministic and may not exceed source-track freshness.

15.2 Support Loss

Soft interruption: temporary radar/FOR/detection dropout enters bounded SUPPORT_HOLDOVER. During holdover the missile propagates its last legitimate estimate. Valid support restored before timeout returns to supported tracking. Timeout expiration commits SUPPORT_LOST. The current target is approximately two seconds; exact value remains Phase-3 gated.

Hard invalidation bypasses holdover and commits SUPPORT_LOST at the next legal support-consumption point. Initial hard causes are defensive-jammer blanking of the supporting organic radar and destruction/unavailability of the support subsystem. Additional hard causes may be added only through the approved Phase-3 contract.

Before autonomous seeker activation, committed SUPPORT_LOST irrecoverably defeats the simplified radar missile. After autonomous activation, launching-radar field of regard and Weapon Support Datalink are no longer required. MEGA LINK / RED LINK cannot substitute for Weapon Support Datalink.

15.3 Active Radar Seeker

When activation criteria are met, the existing missile entity exposes an EmitterView projected from Weapon-owned state with platform_class = MISSILE_SEEKER, radar_role = ACTIVE_SEEKER, and a seeker-specific sensor_profile_id. The seeker does not allocate tactical RadarTrack slots.

The active seeker evaluates a bounded acquisition region around its last legitimate supported estimate using the common radar primitives: FOV, range, target RCS/aspect, authoritative terrain/clutter, Doppler/notch, chaff, jamming, and deterministic acquisition scoring.

On valid acquisition, SensorAndTrack produces an acquired SeekerAssessment; WeaponAndDamage may transition WeaponGuidanceState to TRACKING at the next legal consumption point. Temporary loss produces loss/reacquisition information; WeaponAndDamage owns COASTING_REACQUIRING and its timer. Expiration without reacquisition transitions the weapon to DEFEATED.

An active radar seeker is a valid radar emission and may generate an RWR active-seeker indication according to receiver geometry and sensitivity. An IR seeker remains electromagnetically silent.

16. Stage-15 Processing Order

Internal stage-15 order is deterministic. The following logical order is normative even if implementation helpers are rearranged without changing dependencies:

1. Project persistent radar/link state and Weapon-owned read-only views into stage-15 views.
2. Apply radar, jammer, tactical-link, damage, and capability effects.
3. Generate organic radar observations from truth targets plus eligible chaff using the early-rejection / table-driven detection path.
4. Ingest MEGA LINK / RED LINK reports whose causal/latency boundary has matured; reports denied by current RX state are dropped, not backlog-buffered.
5. Ingest interrogation / IFF / authorized identification evidence.
6. Associate, fuse, and update semantic tactical tracks within each isolated knowledge domain; apply lifecycle, quality, support, and intra-domain capacity rules.
7. Generate outbound MEGA LINK / RED LINK reports from the newly committed eligible semantic tracks; enqueue them for delivery no earlier than the next tick and respect fan-out bounds.
8. Build passive-IR seeker candidate observations from target plus eligible flares.
9. Build active-radar seeker candidate observations from target plus eligible chaff.
10. Evaluate seeker competition and retention hysteresis.
11. Publish final SeekerAssessment records into per-missile SensorHandoffState mailboxes.
12. Produce WeaponSupportUpdate records from newly committed valid local fire-control tracks into per-missile mailboxes.
13. Update RWREmissionObservation -> persistent RWRContactState and MissileThreatState.

17. Phase-3 Gated Data

The engineering contracts above are fixed by v1.0 for rewrite purposes. The following remain authored/measured Phase-3 data rather than fixed numbers in this document:

- radar detection range/aspect tables, scan schedules, beamwidths, PRF modes, clutter/notch thresholds, measurement uncertainty, correlation gates, track coast/promotion thresholds and capacity scoring
- MEGA LINK / RED LINK cadence, transport latency beyond the one-tick minimum, range/geometry/LOS, position quantization, report caps, pending-queue cap, link_tx_priority scoring, external-jam susceptibility, and reacquisition timing
- exact SensorHandoffState record sizes, alignment, mailbox generation/supersession rules, and total byte charge

- aircraft IR-signature tables, multi-engine aggregation, LOW/MIL/MAX engine mapping, IR aspect boundaries, heater sensitivity/range
- flare/chaff signatures, decay, scoring margins, CM maneuver-window filter, and radar/seeker susceptibility
- Weapon Support Datalink cadence, support-holdover timeout, additional hard-invalidation causes, WeaponSupportUpdate layout
- active-seeker activation schedule, FOV/search/acquisition tables, coast/reacquisition rules, countermeasure scoring, retention switch margin/dwell
- exact public record widths, packing, alignment, enum widths, static assertions, and generated registry mappings

18. Acceptance Matrix - Version 1.0 Closure Set

All scenarios are deterministic and tick-indexed. Presentation cadence, PAL/NTSC display service, camera state, and visual LOD must not alter sensor checksums, fire-control outcomes, link state, seeker results, or missile outcomes.

18.1 Organic radar / track / link

- Clear-sky detection and exact PLOT -> TENTATIVE -> TRACK -> WEAPON timing.
- Beam/notch under land and sea clutter; terrain-mask loss/regain at exact ticks.
- 2/4/8-bar revisit behavior and stable observation order.
- Intra-domain 24-track one-over behavior: only eligible nonpriority COAST victim may be evicted.
- Organic + offboard correlation yields one fused track inside a domain and preserves source state.
- MEGA LINK / RED LINK reports obey one-tick minimum causality and bounded fan-out.
- Offboard-only tracks never establish ownership WEAPON quality or SUPPORT_VALID.
- MEGA and RED domains can track the same physical contact independently without cross-domain association, fusion, eviction, or capacity coupling.
- MEGA-domain 24-track load and RED-domain 24-track load can coexist at their legal maxima with no cross-domain resource effect.
- RIO priority remains four MEGA-domain tracks; RED/link transmit priority uses independent semantics.
- Defensive jammer denies MEGA LINK TX/RX; in-flight denied reports are lost, existing offboard tracks coast, and recovery does not replay backlog.

18.2 Seeker / countermeasure / support

- IR acquisition ordering is deterministic across LOW/MIL/MAX and tail -> beam/aft -> forward-quarter -> nose/weak aspects.
- Seeker angle off boresight changes FOV eligibility but not target IR-signature class.
- IR seeker emits no RWR indication; active radar seeker does only when receiver geometry/sensitivity permits.
- Flare/chaff below +3 G consumes inventory and exists but is ineffective; negative 3 G does not qualify.
- Qualified flare/chaff competition follows deterministic geometry/signature scoring and retention hysteresis.
- Beaming alone does not guarantee loss; notch requires geometry + clutter + Doppler + dwell; weaving has no magic bonus.
- Chaff affects airborne/surface/ship/SAM observations before semantic track update and active-seeker competition without allocating tactical tracks.
- SeekerAssessment cannot mutate WeaponGuidanceState until next-tick stage 12 and survives/supersedes through its fixed mailbox rule.
- WeaponSupportUpdate contains no truth-only data and survives the cross-tick handoff without allocation.
- Soft support interruption uses holdover; restoration before timeout prevents defeat; timeout produces SUPPORT_LOST.
- Defensive-jammer / support-subsystem hard failure bypasses holdover at the next legal support-consumption point.
- Active-seeker activation removes launching-radar/weapon-link dependency; MEGA LINK cannot substitute for weapon support.
- Sixteen simultaneous active missile seekers consume zero semantic RadarTrack and RIO-priority slots.
- Twin-engine / engine-out IR combinations map deterministically through the aggregate signature profile.

18.3 Ownership / determinism / capacity

- SensorAndTrack reads Weapon-owned missile/decoy state only through declared read-only views and cannot mutate private WeaponAndDamage state.
- SensorHandoffState bytes, link queue bytes, two 24-track domain stores, and all padding fit the approved SensorTrack owner ledger after rewrite.
- Maximum legal link production fits bounded transport; one-over follows deterministic report-selection/drop policy.
- Entity iteration order changes neither link delivery nor sensor/seeker checksums.
- All above outcomes remain bit-stable across replay seeds when the same sensor RNG seed/state is supplied.

19. Rewrite Propagation Requirements and Next Artefact

Version 1.0 is specifically intended to drive the coordinated document rewrites. The following propagation is required before the rewritten specification set is declared internally aligned:

Document / owner	Required propagation from SensorAndTrack v1.0
Architecture Invariants	Replace ambiguous single 24-track domain semantics with two capacity-isolated 24-track tactical knowledge domains (MEGA/RED); charge the additional 1,920 B base track storage plus handoff/link state to the SensorTrack ledger; retain stage-15 ownership and deterministic pool rules.
Gameplay & Simulation Requirements	Adopt MEGA LINK / RED LINK behavior; offboard-only no ownship WEAPON/support; ownship jammer denies MEGA LINK TX/RX; +3 G CM qualification; simplified IR LOW/MIL/MAX + frontal weak/blind behavior; retain no IR MAWS and geometry-driven defense.
Engine Runtime	Define SensorHandoffState ownership; read-only SeekerInputView/CountermeasureView; SeekerAssessment/WeaponSupportUpdate mailboxes; link queue/fan-out bounds; domain-isolated track stores; corrected stage-15 order.
Read-First / Alignment	List SensorAndTrack v1.0 as the human-reviewed Phase-3 sensor-model baseline and track all parent propagation decisions/hashes.
Generated schema / registry	Freeze exact widths, sizes, alignments, enum mappings, record ownership, mailboxes, link queue, two-domain track representation, and static assertions.

Next artefact: generated logical-interface / schema definition. Numeric radar/IR/seeker tables and alpha-beta filter equations begin only after the rewritten parent contracts and generated schema agree on ownership, capacity, and byte layout.

20. Version 1.0 Human-Review Closure

Human review closes the conceptual SensorAndTrackEngine model at v1.0 for rewrite use. The following decisions are accepted as the baseline to propagate:

- Two capacity-isolated 24-track tactical knowledge domains: MEGA/BUE and RED.
- SensorAndTrack-owned fixed cross-tick SensorHandoffState with explicit byte accounting.
- Independent link transmit priority semantics, not RIO priority reuse.
- Offboard-only data cues/fuses but cannot create ownship weapon-quality/support state.
- Ownship defensive jammer blanks organic radar and denies MEGA LINK transmit and receive while active.
- Simplified deterministic IR signature by aggregate engine state and aspect, including a frontal weak/blind sector.
- Flare and chaff effectiveness require $\geq +3$ G achieved positive normal load in the approved window.
- Common geometry-driven beaming/notching/weaving/chaff/jamming/terrain behavior across air, surface, ship, SAM, and missile radars.
- Current radar missiles use supported midcourse -> autonomous active seeker, not semi-active homing.
- SensorAndTrack produces SeekerAssessment; WeaponAndDamage remains authoritative for seeker-lock/guidance state and missile physics.

End of Version 1.0 - Human-reviewed SensorAndTrackEngine rewrite baseline.